

**DIRE DAWA UNIVERSITY**  
**INSTITUTE OF TECHNOLOGY**  
**SCHOOL OF COMPUTING**  
**DEPARTMENT OF SOFTWARE ENGINEERING**

**INTERNSHIP REPORT**

***“Software Engineering Internship Presentation”***

A report submitted to the Department of Software Engineering, School of Computing, Dire Dawa University Institute of Technology, in partial fulfillment of the requirements of the internship programme for the Degree of BSc in Software Engineering.

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|--------------------------------|---------------------------------|
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*Dire Dawa, Ethiopia*

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## LIST OF ACRONYMS

|                                             |                                                 |
|---------------------------------------------|-------------------------------------------------|
| <b>AI</b> Artificial Intelligence           | <b>API</b> Application Programming Interface    |
| <b>BSc</b> Bachelor of Science              | <b>CBE</b> Commercial Bank of Ethiopia          |
| <b>CRUD</b> Create, Read, Update, Delete    | <b>DBMS</b> Database Management System          |
| <b>DDU</b> Dire Dawa University             | <b>EC</b> Ethiopian Calendar                    |
| <b>ERD</b> Entity-Relationship Diagram      | <b>ETB</b> Ethiopian Birr                       |
| <b>HTML</b> Hypertext Markup Language       | <b>HTTPS</b> Hypertext Transfer Protocol Secure |
| <b>ID</b> Identification                    | <b>IDE</b> Integrated Development Environment   |
| <b>JSON</b> JavaScript Object Notation      | <b>JWT</b> JSON Web Token                       |
| <b>MVC</b> Model-View-Controller            | <b>MVP</b> Minimum Viable Product               |
| <b>OCR</b> Optical Character Recognition    | <b>Odoo</b> Open-source ERP framework           |
| <b>OOP</b> Object-Oriented Programming      | <b>OTP</b> One-Time Password                    |
| <b>ORM</b> Object-Relational Mapping        | <b>PDF</b> Portable Document Format             |
| <b>PWA</b> Progressive Web Application      | <b>QA</b> Quality Assurance                     |
| <b>REST</b> Representational State Transfer | <b>SaaS</b> Software-as-a-Service               |
| <b>SDLC</b> Software Development Life Cycle | <b>SQL</b> Structured Query Language            |
| <b>TOTP</b> Time-based One-Time Password    | <b>UI</b> User Interface                        |
| <b>URL</b> Uniform Resource Locator         | <b>UX</b> User Experience                       |
| <b>VCS</b> Version Control System           |                                                 |

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## **5. INTRODUCTION**

### **5.1 Purpose of the Internship**

The purpose of my internship was to acquire practical software engineering experience by participating in real projects, collaborating with developers and fellow interns, and applying academic knowledge to user-facing and business-oriented systems. I aimed to understand the complete delivery process: studying an existing codebase, interpreting requirements, implementing within an assigned boundary, connecting layers, testing, presenting, revising, and integrating. The internship programme at Dire Dawa University is structured as a full-semester industrial placement completed after the third year of study, designed to bridge the gap between classroom learning and professional practice by exposing students to real client requirements, real deadlines, real code review, and real production considerations. For me specifically, the programme aligned with my personal interests in web development and artificial intelligence, both of which I had the opportunity to explore in depth during the four months at Afronex.

The primary goal was not merely to write code but to understand the full software development life cycle in a professional context: how requirements are interpreted from client communications, how features are planned and assigned across teams, how code is reviewed and integrated, how testing and quality assurance validate the work, and how presentations and feedback drive iterative improvement. I also sought to develop the communication and leadership skills necessary to coordinate across teams and to serve as an effective bridge between the intern cohort and the company leadership.

### **5.2 How It Relates to Your Programme of Study**

We applied most skills that we acquired through the three years of learning software engineering at DDU in order to solve real-world problems. The internship fits the sectors I want to specialise upon, particularly web development and AI. Over the preceding three years, I completed courses in Object-Oriented Programming, Data Structures and Algorithms, Object-Oriented System Analysis and Design, Advanced Software Engineering, Fundamentals of Database Systems, Software Architecture and Design, Software Security, Internet Programming, Mobile Application Development, and Software Engineering Tools and Practices, among others as per the DDU curriculum. During the internship, I drew on these foundations daily: Object-Oriented Analysis informed the way I

read existing class diagrams; Advanced Database Systems guided my work on the payment audit trail; Software Architecture shaped my understanding of the seven-team allocation structure; and Software Security informed my approach to payment screenshot handling and OTP confirmation flows.

### **5.3 Personal Learning Objectives**

My personal learning objectives were to learn anything useful for the project we were working on, to strengthen my full-stack web development skills (not limited to frontend), to gain practical exposure to payment and subscription workflows, to improve my debugging and testing habits, and to explore the responsible use of AI-assisted engineering. I also wanted to understand how a complex product is divided among task-based teams and how cross-team integration works in practice. An additional objective was to learn how to monitor and direct agentic AI systems when performing tasks outside my primary expertise, which shaped my preference for the Django framework (Python-based, compatible with the AI ecosystem) on selected personal and company-assigned projects.

### **5.4 Initial Assumptions and How They Changed**

I assumed building the whole system would be given to me to do as the company wanted, and to be integrated and deployed directly into production. This assumption made me nervous, as I feared any mistake would have immediate real-world consequences. The reality was very different: the work was fundamentally about teamwork, with each intern assigned to a specific role based on their strengths, and integration happened only after multiple rounds of review, testing, and cross-team coordination. I learned that production readiness is not the responsibility of a single developer but the collective outcome of a disciplined team process: shared contracts, code review, integration testing, acceptance criteria, and stakeholder sign-off. I also learned that asking a precise question early was often more efficient than implementing an assumption and discovering later that it conflicted with another module's expectations.

### **5.5 Planned Schedule, Onboarding, and First Assignment**

I made myself busy from the first day with learning and implementing until I reached what was expected of me. There was no rigid schedule that framed my working days; instead, we were advised to manage our own time while ensuring project milestones were met. The company provided a final submission date and a presentation day for each project, on which

we would demonstrate the work, check integration issues, and fix defects. Onboarding consisted of reading shared documentation including `developer-prompt.md`, `team-ownership-matrix.md`, `delivery-roadmap.md`, and the `team-allocation` file. My first assignment was to locate my teams, read the feature plans and checklists, and keep the `developer-prompt.md` file open as the authoritative reference. We were supervised through weekly meetings with no fixed dates, scheduled around project progress.

## 6. COMPANY BACKGROUND

### 6.1 Organizational Overview

Afronex Tech Company is found in Dire Dawa city, Ethiopia, near Dire Dawa University. The company is known for delivering real-time technology solutions for real-world problems across multiple domains including system development, web development, artificial intelligence, and technology skills development. Its client base spans the eastern region and the wider national market, with a focus on serving government institutions, membership-based organisations, and small-to-medium enterprises. Afronex is led by Mr. Tsegaw, who is also a lecturer at DDU. His academic background in software engineering informs the company's working culture, which combines discipline with mentorship. He was friendly with his students and continually encouraged us to advance on software development and AI usage. The company is now focusing its strategic direction on AI-driven products. Afronex also runs Afronex Tech Hub, a skills-development initiative that teaches practical software engineering to early-career developers. This culture of openness made the internship particularly valuable: interns were integrated into active project teams, given ownership of specific features, and held accountable for delivery in the same way as full-time engineers.

### 6.2 Services and Technology Orientation

The internship environment reflected the needs of modern digital services: accessible interfaces, responsive workflows, secure handling of sensitive data, integration between modules, and adaptability to changing requirements. Afronex's service portfolio spans custom web application development, membership management systems, AI-driven care solutions, accounting and inventory subsystems built on Odoo, and a growing set of AI product initiatives. Because the projects involved confidential organisational and client information, all references in this report have been generalised. The company's technology orientation favours mature, well-supported frameworks: React with TypeScript for frontend work on the main project, Django and Odoo for Python-based backend work on other projects, and agentic AI patterns for selected development activities. It should be noted that there was no fixed tech stack that obligated me to learn one specific technology for all projects: the membership management system was assigned to the intern team with a specified tech stack (React, TypeScript, PostgreSQL, Prisma ORM), while other projects I

participated in used tech stacks chosen by my or the company's preference (Django for AI compatibility, Odoo for ERP workflows).

### 6.3 Organizational Structure and Intern Placement

Interns were assigned across all project groups according to the tasks and ownership areas defined for the product. I participated in almost all parts of the membership management system, but to specify my primary role I preferred the Payments and Subscriptions feature (Team 6), and I also contributed boldly to the Super Admin and Organization Governance feature (Team 2). The full team structure comprised seven teams, each with a clearly defined ownership area, as summarised in Table 6.1. The assignment order followed a strict dependency chain: Team 1 started first to lock shared platform contracts; Teams 2 and 3 started after Team 1's contracts were stable; Teams 4, 5, and 6 started after shared API and status rules were finalised; and Team 7 ran across the integration phase to validate cross-module behaviour.

Table 6.1: Team Allocation for the Membership Management System Modernisation

| Team            | Role                                 | Key Responsibilities                                                                         |
|-----------------|--------------------------------------|----------------------------------------------------------------------------------------------|
| <b>Team 1</b>   | Platform Foundation and Security     | Auth, roles, permissions, API response format, error handling                                |
| <b>Team 2 ★</b> | Super Admin and Org Governance       | Super admin dashboard, organizations management, org admin oversight, platform-level control |
| <b>Team 3</b>   | Org Admin and Member Management      | Member CRUD, import/export, bulk actions                                                     |
| <b>Team 4</b>   | Event Management and Registration    | Event CRUD, calendar views, attendee management                                              |
| <b>Team 5</b>   | Blog, Announcements, Content         | Post CRUD, draft/publish workflow, filters                                                   |
| <b>Team 6 ★</b> | Payments and Subscriptions           | Payment transactions, subscription plans, invoices, receipts, payment status normalization   |
| <b>Team 7</b>   | Member Experience and Integration QA | Cross-module validation, regression catching, release-readiness                              |

### 6.4 Working Culture

The working culture emphasised communication, peer support, iterative improvement, and practical completion. Work was organised around expected deliverables, demonstrations, and integration milestones rather than rigid daily stand-ups. We were encouraged to come forward with new feature recommendations on the project and on the working environment.

Progress was demonstrated through presentations to the company leadership and to fellow interns, after which comments were collected, prioritised, and incorporated into the next iteration. Several recommendations I made were subsequently adopted, as described in Section 12.

## 7. INTERNSHIP ACTIVITIES

### 7.1 My Role and Responsibilities

My role during the internship was not confined to a single layer or a single team. I worked as a full-stack developer with cross-cutting responsibilities spanning frontend implementation, backend integration, API design, database schema contributions, and AI-assisted engineering. I was also designated as the communication person on the interns' side, serving as the primary point of contact between the intern cohort and the company leadership, coordinating cross-team discussions, and preparing progress presentations.

My primary team assignment was Team 6 (Payments and Subscriptions), where I owned payment transactions, subscription plans and billing state, manual payment and transfer recording, invoices and receipts workspace, and payment status normalization and auditability. In addition to Team 6, I contributed boldly to Team 2 (Super Admin and Organization Governance), where I worked on the super admin dashboard, organizations management, org admin oversight, and platform-level control. My participation across both teams gave me visibility into how governance and payment systems interact, which proved valuable during the integration phase when Teams 2 and 6 needed to coordinate on organisation-level billing rules.

Beyond the main project, I participated as a prompt engineer and AI engineer on selected tasks, deploying agentic AI systems to perform scoped activities for areas where our group was not yet advanced. I was also tasked with coming up with new recommendation features on the project and on the company's running products. A summary of my key activities and outcomes is presented in Table 7.1.

Table 7.1: Summary of Internship Activities and Outcomes

| Area                                            | My Practical Activities                                                                                                 | Evidence / Outcome                                         |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Full-stack Development (Teams 2 &amp; 6)</b> | Implemented frontend + backend for payment confirmation, super admin dashboard, org governance, invoice workspace       | Feature implementations, integration tests, demonstrations |
| <b>Team 6: Payments &amp; Subscriptions</b>     | Owned payment transactions, subscription states, manual payment confirmation, invoices, receipts, payment normalization | Primary task ownership, cross-module integration           |
| <b>Team 2: Super Admin &amp; Org Governance</b> | Contributed to super admin dashboard, organizations management, org admin oversight, platform-level control             | Bold contribution to governance module                     |

| Area                                       | My Practical Activities                                                         | Evidence / Outcome                                                |
|--------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------------|
| <b>Prompt Engineering / AI Engineering</b> | Deployed and monitored agentic AI for scoped tasks outside group's expertise    | AI-assisted scaffolding, test generation, security review         |
| <b>Communication Lead (Interns)</b>        | Primary contact between interns and company; coordinated cross-team discussions | Progress presentations, issue escalation, cross-team coordination |
| <b>Feature Recommendations</b>             | Proposed new features for projects and company products                         | Several recommendations adopted (see Section 12)                  |
| <b>Personal Projects</b>                   | Developed Temariware (Telegram Mini App) outside company hours                  | Project now live in production                                    |

## 7.2 Time Distribution

Over the course of the internship, my time was distributed as follows: development 40%, documentation and meetings 30%, testing 10%, research 5%, and support/training/other 15%. Development occupied the largest share, reflecting the depth of full-stack and integration work required across Teams 2 and 6. Documentation and meetings together accounted for 30%, reflecting the company's strong emphasis on shared contracts, plan documents, checklists, and weekly review sessions. The relatively low testing percentage (10%) reflects the fact that formal testing was coordinated by Team 7 (Integration QA), while individual teams focused on implementation and manual verification of their own features.

## 7.3 Main Project: Membership Management System Modernisation

The principal team project was the modernisation of a previously existing membership management system owned by Afronex. Our task as a team of interns was to rewrite the system using modern frameworks with additional features requested by the client. For confidentiality, the client name and production URLs cannot be disclosed. The modernisation included the following key updates: (1) Fayda ID card integration as a primary identification mechanism for member registration; (2) custom attributes for member profiles, allowing each organisation to define its own fields; (3) Telebirr payment integration for membership fees and subscription renewals; (4) manual payment confirmation and invoice integration for cases where automated verification was unavailable; (5) OTP confirmation for user registration to verify phone number ownership; and (6) OCR-based payment screenshot verification, allowing users to upload a screenshot of their Telebirr or CBE Birr success screen and have the system extract the transaction ID,

amount, and date using an OCR library. This approach significantly reduced user friction compared with manual entry of long transaction strings.

Within Team 6, my specific contributions focused on the manual payment confirmation workflow, the invoice and receipt workspace, and the integration of the OCR-based screenshot verification flow. I implemented both the frontend interfaces and the backend integration logic for the manual payment confirmation flow, and I contributed to the design of payment status normalization rules, ensuring that every payment record regardless of source was represented in a consistent state machine that could be audited end-to-end. Within Team 2, I contributed to the super admin dashboard, implementing organisation management interfaces, org admin oversight views, and platform-level control features that allowed administrators to manage the status and visibility of organisations across the system.

A particular area of independent contribution was the OCR-based payment screenshot verification flow. The user-facing concept was simple: instead of asking users to type long transaction IDs, they would upload a screenshot of their Telebirr or CBE Birr success screen, and an OCR library (such as Tesseract or EasyOCR) would extract the transaction ID, amount, and date from the image. This approach had very low friction for the user, but it required careful design on the backend side: the OCR output had to be validated against expected formats, the extracted transaction had to be checked for uniqueness to prevent double-processing, and the screenshot itself had to be retained as an audit trail for the manual confirmation step. I worked on the full-stack integration of this flow with the manual payment confirmation workspace owned by Team 6.

## **7.4 Other Projects Participated**

Beyond the main team project, I participated in several other initiatives at Afronex. I contributed to an AI-driven care system built using the Django framework, selected for Python because of its strong AI ecosystem compatibility. My role involved contributing to the system's data flow design and integrating selected AI-driven features; the specific client and domain cannot be disclosed for confidentiality. I also participated in the development of an accounting and inventory subsystem for another of the company's main projects using Odoo (an open-source ERP framework in Python), focusing on adapting existing modules to meet client-specific workflow requirements including custom report generation and integration with the broader system's payment records. For both of these projects, the technology stack was chosen based on my or the company's preference, not as a fixed

requirement — Django was preferred for its Python/AI ecosystem compatibility, and Odoo was chosen for its modular ERP architecture that fit the accounting domain.

Additionally, I developed a personal production-grade project outside company hours: Temariware, a Telegram Mini App that connects Ethiopian university students with verified tutoring, freelance, and part-time job opportunities. Temariware is built on Next.js, React, TypeScript, Prisma ORM, and Neon Postgres, deployed on Vercel's free tier, with Telegram WebApp initData HMAC-SHA256 authentication and screenshot-based payment verification. The system is live and accepting real users, and it directly applies the skills and perspectives I gained during the internship — particularly the payment verification flow design and the business-logic mindset shift. I also explored other personal project ideas in the agritech space, applying the same full-stack and AI engineering skills to different problem domains.

## **7.5 Tools, Programming Languages, and Frameworks Used**

The tools, languages, and frameworks used during the internship were determined by the project context. For the membership management system modernisation (the main team project), the company specified the tech stack: React with TypeScript for the frontend, PostgreSQL via Prisma ORM for the database, and Git/GitHub for version control with structured pull request review. For the AI care system, Django (Python) was used, chosen for its AI ecosystem compatibility. For the accounting and inventory subsystem, Odoo (Python) was used, chosen for its modular ERP architecture. For my personal project Temariware, I used Next.js, React, TypeScript, Tailwind CSS, shadcn/ui, Prisma ORM, and Neon Postgres, chosen based on my preference for modern full-stack development. Additional tools included: agentic AI systems for code scaffolding and security review, Tesseract and EasyOCR for OCR, Visual Studio Code as the primary IDE, Postman for API testing, Telegram groups and Zoom meetings for collaboration, and collaborative folders for AI agent outputs. It should be noted that there was no fixed tech stack obligation: I learned the frameworks the company specified for the membership management system, and for other projects I used frameworks chosen by my or the company's preference based on the problem domain.

## **7.6 Approach and Processes Followed**

The company used an Agile development methodology for the membership management system modernisation. The Agile approach was chosen because the project requirements

evolved as the client provided feedback on early iterations, and a traditional waterfall approach would have locked in decisions too early. Work was organised into sprints aligned with feature ownership: each team owned a set of features and was responsible for delivering them within an agreed timeline, with integration and review happening at the end of each sprint. The per-feature workflow was: read the plan document, read the checklist, implement on a dedicated branch, submit a pull request for review, undergo integration testing by Team 7, validate against acceptance criteria, and present to the company for feedback. The team-allocation structure (seven teams with a strict dependency order) was itself a manifestation of the Agile principle of organising work around autonomous, cross-functional teams.

## **7.7 Examples of Problems Solved**

Several concrete problems were solved during the internship, three of which are highlighted here. First, the payment status normalization problem: payments arrived from multiple sources (Telebirr, manual transfer, OCR-verified screenshot), each with its own data shape and verification status. The solution was a unified state machine that mapped every payment source to a common status set (PENDING, APPROVED, REJECTED) with an audit trail that recorded who reviewed each payment and when. Second, the organisation governance problem (Team 2): the super admin needed to manage organisations across the platform with different status actions and summaries, while org admins needed oversight limited to their own organisation. The solution was a role-based access control layer that distinguished between super admin, org admin, and member roles, with platform-level visibility for the super admin and org-level visibility for org admins.

Third, the OCR accuracy problem: the OCR-based payment screenshot verification flow initially produced inconsistent results due to variations in screenshot quality, layout, and language. The solution was a multi-step validation pipeline: the OCR output was parsed for the expected fields using regular expressions tuned to Telebirr and CBE Birr formats, the extracted values were validated for uniqueness against existing payment records to prevent double-processing, and any uncertain extraction was routed to the manual confirmation workspace for human review. This combination of automated extraction and human fallback achieved an acceptable balance between user friction and verification reliability.

## **8. TECHNICAL SKILLS AND KNOWLEDGE UTILIZED**

### **8.1 Programming Languages and Frameworks**

During the internship, I worked with several programming languages and frameworks, each selected for a specific project context. TypeScript was the primary language for the membership management system modernisation, chosen by the company for its static typing which reduced runtime errors. Python was used for the AI care system (via Django) and the accounting and inventory subsystem (via Odoo), taking advantage of Python's AI ecosystem compatibility. SQL was used for database queries with Prisma ORM abstracting routine SQL. It should be noted that there was no fixed tech stack that obligated me to learn one specific technology: the company specified React and TypeScript for the membership management system, while for other projects (AI Care, Odoo accounting, and my personal project Temariware), the tech stack was chosen by my or the company's preference based on the problem domain and AI ecosystem compatibility.

Frameworks used included: React with TypeScript (primary frontend for the modernisation, company-specified), Next.js (for personal projects), Django (for the AI care system, preference-based for Python/AI compatibility), Odoo (for the accounting subsystem, preference-based for ERP workflows), Tailwind CSS with shadcn/ui (styling), and Prisma ORM (database management for TypeScript-based work). The diversity of frameworks reflected the reality of professional software engineering: the right tool depends on the project context, not on a single developer's preference.

### **8.2 Tools, Version Control, and Testing**

Git and GitHub were used throughout for version control, with structured branching conventions and pull request review documented in the shared developer-prompt.md file. Each feature was developed on a dedicated branch, reviewed by at least one team member, and integrated only after passing the relevant checklist. Other tools included Visual Studio Code as the primary IDE, Postman for API testing, and LibreOffice for documentation. For collaboration, the team used Telegram groups for asynchronous communication, Zoom meetings for synchronous discussions, collaborative folders for AI agent outputs, and phone calls for urgent coordination. Testing included manual verification of payment flows, integration testing across module boundaries (coordinated by Team 7), and acceptance criteria validation before features were considered complete. For my personal project, I

implemented more structured testing using the React Testing Library for component tests and end-to-end smoke tests for critical flows.

### 8.3 Soft Skills

Teamwork was the most significant soft skill developed: I learned how to coordinate with cross-functional teams across two team assignments (Team 2 and Team 6), communicate delays early, and accept that my preferred implementation might not always be the right one for the broader system. Communication was developed through my role as the company's communication lead on the interns' side, which required translating between technical and non-technical stakeholders, preparing progress presentations, and facilitating cross-team discussions. Time management was developed through the company's flexible working culture, which required me to set my own priorities and meet deadlines without rigid supervision. Leadership was developed unexpectedly: as the communication lead, I had to coordinate cross-team discussions, facilitate disagreement resolution, and ensure that every team's voice was heard in the broader planning process.

### 8.4 Coursework Connections

The internship directly connected to my university coursework at DDU. Table 8.1 maps key courses to their practical applications during the internship, illustrating the strong alignment between the DDU Software Engineering curriculum and professional software development.

Table 8.1: Mapping of University Coursework to Internship Applications

| University Course                                 | Internship Application                                                             |
|---------------------------------------------------|------------------------------------------------------------------------------------|
| <b>Object-Oriented Programming</b>                | Reading and refining existing class diagrams for the payment state machine         |
| <b>Object-Oriented System Analysis and Design</b> | Use case modelling for the manual payment confirmation and org governance flows    |
| <b>Advanced Software Engineering</b>              | Understanding the seven-team allocation; iterative development with weekly review  |
| <b>Fundamentals / Advanced Database Systems</b>   | Payment audit trail design; subscription state machine; status normalization       |
| <b>Internet Programming</b>                       | React frontend implementation; Next.js API route development; REST API consumption |
| <b>Software Architecture and Design</b>           | Understanding the layered architecture; component and deployment diagrams          |
| <b>Software Security</b>                          | Payment screenshot handling; OTP confirmation flow; audit trail design             |
| <b>Software Engineering Tools and Practices</b>   | Git and GitHub workflow; pull request review; shared markdown documentation        |

## **8.5 Certifications and Training**

There were no formal certifications issued during the internship itself. However, I participated in the Cursor Hackathon held at Haramaya University, which brought together participants from multiple universities. I received a certificate of participation for this event, which is included in the appendices. Beyond the hackathon, training was informal but continuous: weekly supervision meetings, code review feedback, and presentation review sessions all served as ongoing professional development throughout the four months.

## **9. COLLABORATION AND TEAMWORK**

### **9.1 Team Experience and Collaborative Tools**

My experience working with the team at Afronex was one of the most formative aspects of the internship. The team comprised classmates from DDU and students from other universities including Haramaya University. We were organised into pairs within each team, with each pair taking ownership of a specific feature based on preference and closeness. My dual assignment to Team 2 (Super Admin and Org Governance) and Team 6 (Payments and Subscriptions) gave me visibility into how governance and payment systems interact, which proved valuable during the integration phase when Teams 2 and 6 needed to coordinate on organisation-level billing rules. Collaborative tools used included: Telegram groups for asynchronous communication, GitHub for code hosting and pull request review, collaborative folders for AI agent outputs, physical discussions at the company premises, phone calls for urgent coordination, and Zoom meetings for synchronous cross-university discussions.

### **9.2 Achievements and Contributions**

Through the team experience, I gained significant experience in teamwork, solving real-world problems, advancing my use of trending technologies (particularly agentic AI), and following the right software development life cycle from requirements through to production. A specific achievement was my selection, due to my commitment and communication skills, to participate in and contribute to real client-facing projects of the company that are still in development. This opportunity was extended based on the trust I built through my dual-team work (Teams 2 and 6) and through my role as the communication lead on the interns' side. I am continuing to collaborate on these projects beyond the formal end of the internship.

### **9.3 Disagreements, Misunderstandings, and Coordination Problems**

A recurring challenge was that sometimes one of us would try to dominate the team's flow and insist on a single direction, creating disagreements among teammates. We resolved this through discussion, sometimes repeatedly, until we reached a shared understanding. If the disagreement was too far apart to be resolved internally, we would escalate to the company leadership to guide us on the best alternative. Through this process, we learned to value team spirit over individual preference, to separate the person from the technical position,

and to make decisions based on evidence and shared contracts rather than on force of personality.

#### **9.4 Time Management, Feedback, and Professional Behaviour**

My approach to time management was to first learn the unfamiliar aspects of a task, then work on the project keeping priority, time management, and punctuality in focus. The company's flexible working culture supported this approach as long as deliverables met agreed deadlines. I was appreciated for my activeness in utilising new technology advancements, particularly AI, and after receiving this feedback I now plan to specialise in AI engineering for my future career. Professional behaviours developed included politeness, team spirit, communication, and team leadership. As the communication lead on the interns' side, I had to coordinate cross-team discussions, facilitate disagreement resolution, and ensure that every team's voice was heard in the broader planning process.

## 10. PROJECT HIGHLIGHTS

### 10.1 Overview

During the internship period, I worked on three significant projects: the membership management system modernisation at Afronex (the main company project, described in Section 7 with confidentiality constraints), the AI care system built in Django (described in Section 7.4), and a personal production-grade project built outside company hours: Temariware. The personal project is now live in production and represents the application of internship-learned skills to a problem I am personally passionate about. The company projects cannot be described in full detail due to confidentiality; the personal project is highlighted below.

### 10.2 Main Project: Membership Management System Modernisation

The membership management system modernisation was the principal team project at Afronex. The problem: an existing membership management system built on an earlier framework had accumulated technical debt that limited extensibility, lacked digital ID integration (Fayda), had no Telebirr payment support, used a rigid schema with no custom attributes, had no OTP verification during registration, and had no screenshot-based payment verification for user convenience. My role: full-stack developer with primary ownership of Team 6 (Payments and Subscriptions) and bold contribution to Team 2 (Super Admin and Org Governance). Technologies used: React with TypeScript (company-specified frontend), PostgreSQL via Prisma ORM (company-specified database), Git and GitHub (version control), Tesseract and EasyOCR (for the OCR screenshot verification), and agentic AI systems for selected scaffolding and review tasks.

Challenges faced and overcome: the payment status normalization problem was solved with a unified state machine mapping all payment sources to common statuses; the OCR accuracy problem was solved with a multi-step validation pipeline combining automated extraction with human fallback; and the organisation governance problem (Team 2) was solved with a role-based access control layer distinguishing super admin, org admin, and member roles. All features passed integration testing by Team 7 and are being tested for production deployment by the company. The code I wrote is now being reviewed for production, which validates the quality of the full-stack work I delivered across both teams.

### 10.3 Personal Project: Temariware

Temariware is a Telegram Mini App combined with a Telegram Bot that connects Ethiopian university students with verified tutoring, freelance, part-time, and full-time job opportunities. The problem it addresses is that informal Telegram job channels post unstructured job ads with no filtering, no archive, no verification, and no payment mechanism. Temariware provides a structured, multilingual platform (English, Amharic, Afaan Oromoo) where students can browse verified jobs, apply with their academic profile including mandatory documents (National ID, University ID, grade report), and pay a confirmation fee through a screenshot-based verification flow — directly inspired by the OCR-based payment verification I implemented at Afronex. Employers can post jobs with rich metadata, set per-job application fees, review applicant documents in a dedicated panel, and have the platform automatically close positions once hiring targets are met. The bot (@temariwarebot on Telegram) exposes nine slash commands and pushes instant job alerts to subscribers, while the @TEMARIWARE channel automatically receives richly-formatted job cards every time an administrator approves a new posting. The complete stack is built on free-tier infrastructure: Next.js 16 + React 19 + TypeScript + Tailwind CSS + shadcn/ui, Prisma ORM + PostgreSQL on Neon, Vercel hosting, and Telegram WebApp initData HMAC-SHA256 validation. Total monthly cost: 0 ETB. The system is live and accepting real users. Temariware is a solo project in which I am the sole developer and architect, and it represents the direct application of skills and perspectives gained during the internship — particularly the payment verification flow design and the business-logic mindset shift.

### 10.4 Other Project: AI Care System (Django)

I also contributed to an AI-driven care system built using the Django framework, selected for Python because of its strong AI ecosystem compatibility. My role involved contributing to the system's data flow design and integrating selected AI-driven features. The specific client and domain cannot be disclosed for confidentiality. The technology stack was chosen by preference (not company-mandated): Django was selected for its Python/AI ecosystem compatibility, which aligned with my personal interest in artificial intelligence and my role as an AI engineer on selected tasks. The challenges faced on this project included designing data flows that could accommodate both structured and unstructured AI-generated content, and ensuring that the AI-driven features maintained appropriate human oversight — a

pattern I had already established during the membership management system work with agentic AI systems.

## **11. LEARNING AND GROWTH**

### **11.1 New Skills Acquired**

During the four-month internship, I acquired a broad range of new skills spanning technical, methodological, and professional dimensions. Technically, I deepened my proficiency in React and TypeScript (company-specified for the main project), learned the Django and Odoo frameworks for Python-based web development (preference-based for other projects), became familiar with the Telegram Bot API and Mini App framework, and gained practical experience with Prisma ORM and PostgreSQL. I also developed practical skills in AI-assisted engineering: prompt engineering for code scaffolding, agentic AI deployment for scoped tasks, and the use of OCR libraries for payment verification. Methodologically, I learned how an Agile development process works in a professional environment: iterative development with weekly review, shared contracts across teams, dependency-ordered team allocation, and structured pull request review. Professionally, I developed communication and leadership skills through my role as the company's communication lead, time management skills through the company's flexible working culture, and cross-team coordination skills through my dual assignment to Teams 2 and 6.

### **11.2 The Shift from Code Correctness to Business Logic Thinking**

The most significant intellectual growth during the internship was a shift in my mindset from asking “is the system running without errors?” to asking “how can I generate income or build the business logic behind each feature of the project?” Working for the business logic behind each feature required the most guidance for me. I focused on “is the system running with no error?” but not on “how can I generate income or build the business logic behind each feature?” After the guidance I received from Mr. Tsegaw and the senior developers, I am now somewhat business-minded while coding, and I evaluate technical decisions not only for their correctness but also for their contribution to the business. This shift is, in my view, the single most valuable learning outcome of the internship.

### **11.3 Independent Ability Demonstrated**

Three tasks best demonstrate my independent ability. First, researching and advancing features: when I encountered an unfamiliar feature area, I would independently research the available options, propose an implementation approach, and advance the feature through review. Second, developing and monitoring agentic AI to perform specific tasks: I deployed

AI agents for scoped activities such as code scaffolding and security review, monitored their output, and integrated validated results into the team's work. Third, checking for security issues like confidentiality: I developed a habit of reviewing sensitive flows, particularly around payment handling and member data, for potential confidentiality breaches and proposing mitigations.

#### **11.4 Hackathon Participation and Continuous Learning**

I participated in the Cursor Hackathon held at Haramaya University, which brought together participants from multiple universities. I received a certificate of participation (see Appendix B). The hackathon reinforced lessons from the internship: the value of clear communication under time pressure, the importance of prioritising features that deliver value quickly, and the satisfaction of building something that works end-to-end within a constrained timeframe. Beyond the hackathon, learning was continuous through weekly supervision meetings, code review feedback, and self-directed learning in agentic AI patterns, which directly informed my personal project Temariware.

## **12. CHALLENGES FACED AND SOLUTIONS**

### **12.1 Main Difficulties Encountered**

The most significant technical challenge was working with frameworks that the company wanted for the main project, as I had previous experience with different frameworks for the same problem domain. My personal preference for Django (because of Python's AI ecosystem compatibility) was different from the company's specification of React and TypeScript for the membership management system. Bridging this gap required investing additional time in learning the company's preferred stack to a production-ready level, while preserving my Django expertise for other projects where it was the right tool. Other technical challenges included: understanding large existing codebases with limited documentation; integrating features across module boundaries when shared contracts were still in flux; debugging issues spanning frontend, backend, and database layers; and handling the inherent uncertainty of AI agent outputs when using them for code scaffolding and security review.

On the professional side, the most significant challenge was the shift from a purely technical mindset to a business-aware mindset. Working for the business logic behind each feature required the most guidance for me, because my academic training had focused almost exclusively on the technical correctness of code rather than on its commercial viability. Additional challenges included managing communication across cross-functional teams with different working styles, handling disagreements constructively without dominating the discussion, and balancing the depth of feature ownership across two teams (Team 2 and Team 6) with the breadth of cross-team awareness required for effective integration.

### **12.2 How I Overcame Them and What I Learned**

I overcame these challenges through self-directed learning, structured mentorship, and disciplined practice. When I encountered an unfamiliar framework, I trained myself using resources shared by the company. I made a habit of reading the existing codebase before writing any new code. When I encountered integration challenges, I escalated with a clear description of the problem and the options I had already considered, which allowed senior developers to provide targeted guidance. For AI agent outputs, I developed a validation pipeline: every AI-generated artefact was reviewed by a human team member, checked

against project conventions, and only integrated after passing both automated and manual review.

The challenges taught me several durable lessons: the right framework depends on the project context, not on developer preference; integration is the hardest part of multi-team development; business awareness is not optional for a software engineer; AI agents are powerful tools but not a substitute for engineering judgement; and communication is a first-class engineering skill. An additional risk-management decision was choosing to remain in Dire Dawa and join Afronex rather than travelling to Addis Ababa as the school advised; in retrospect, this decision was correct given the quality of mentorship and project work at Afronex.

## **13. CONTRIBUTION TO THE ORGANIZATION**

### **13.1 Recommended Features Now in Use**

Several recommended features I proposed during the internship, both on the projects we were working on and on the way the company acts, are now in use by the organisation. These recommendations ranged from specific feature ideas for the membership management system to broader suggestions for how the company could integrate AI more deeply into its development workflow. The fact that these recommendations were adopted and are now being tested for production validates the value of the internship programme not only as a learning experience for the intern but also as a source of fresh perspectives for the host company. Specific examples, while kept general for confidentiality, include improvements to the manual payment confirmation workflow that reduced review time, refinements to the OCR-based screenshot verification pipeline that improved extraction accuracy, suggestions for how agentic AI could be deployed for selected development activities while preserving human oversight, and improvements to the organisation governance interfaces that streamlined the super admin workflow.

### **13.2 Production-Ready Code and Deliverables**

The code I wrote during the internship, particularly within Team 6 (Payments and Subscriptions) and Team 2 (Super Admin and Org Governance), is now being tested by the company for production deployment. Deliverables included: the full-stack implementation of the manual payment confirmation workflow (frontend + backend); the integration code for the OCR-based screenshot verification pipeline; the design and implementation of payment status normalization rules; the super admin dashboard interfaces for organisation management and org admin oversight; documentation of the payment audit trail; and plan and checklist documents reusable by future interns. The code followed the company's established conventions for branching, review, and integration, and it passed the acceptance criteria defined by Team 7 during the integration phase. For my commitment and communication, I was given the opportunity to participate in real client-facing projects that are still in development, and I am continuing to collaborate on these beyond the formal end of the internship, which represents an extension of the trust the company placed in me based on my performance across both teams.

## **14. CONCLUSION & RECOMMENDATIONS**

### **14.1 Summary of Overall Experience**

The four-month internship at Afronex Tech Company was a transformative experience that bridged the gap between my academic training at DDU and the practical demands of professional software engineering. I worked on a real client-facing project (the membership management system modernisation) as a full-stack developer with primary ownership of Team 6 (Payments and Subscriptions) and bold contribution to Team 2 (Super Admin and Org Governance). I contributed to additional company projects (an AI care system in Django and an accounting subsystem in Odoo), developed a personal production-grade project (Temariware) that is now live in production, participated in a hackathon at Haramaya University, and built durable professional relationships. The most significant learning outcomes were: the shift from a purely technical mindset to a business-aware mindset; the development of communication and leadership skills; practical experience of working within an Agile, multi-team development process across two teams; and the confidence that comes from having one's code tested for production deployment.

### **14.2 Suggestions for Improving Future Internship Programmes**

Based on my experience, I offer the following suggestions. First, the university could provide more structured guidance on company selection, including a curated list of partner companies with documented track records. Second, a formal feedback loop between interns and host companies could be established, collecting mid-internship and end-of-internship feedback from both sides. Third, the host company could formalise the onboarding process by providing a written guide to conventions, tools, and expectations on the first day. Fourth, both the university and the host company could provide more structured opportunities for cross-cohort networking, so that interns can learn from the experiences of those who came before them.

### **14.3 Recommendations for Future Interns**

For future interns, I recommend: (1) do not wait to be told what to do — take ownership of your learning from day one; (2) balance depth with breadth — master the features you own but also understand how they connect to the broader system; (3) cultivate a business-aware mindset from the beginning — ask why each feature matters, not just how to implement it; (4) embrace AI as a tool but do not abdicate engineering judgement to it; (5) communicate

proactively — if you are stuck, say so early; if you see a problem, raise it with a proposed solution; (6) document your work as if a future intern will need to understand it without your help.

#### **14.4 How This Internship Will Impact My Future Career**

I had very important communication with the company and with others, and I gained practical experience of working in teams on real-world problems. I am sure that something interesting is coming in the future. The internship has shaped my career direction in three ways: it has confirmed my interest in AI engineering, which I now plan to pursue as my primary specialisation; it has given me the confidence that I can build production-grade software from scratch, as demonstrated by my live personal project Temariware; and it has given me a professional network that I will continue to draw on. In the long term, I aspire to build a technology company of my own in Ethiopia, applying the lessons learned at Afronex about combining technical depth, business awareness, and a culture of mentorship.

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## 16. APPENDICES

### Appendix A: Weekly Log Summary

This appendix summarises the weekly activities during the four-month internship period.

Table A.1: Weekly Activity Log Summary

| Period                     | Key Activities and Milestones                                                                                                                              |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Month 1 (Sene 2018 EC)     | Onboarding, reading shared documentation, joining Teams 2 and 6, first feature assignments, learning React + TypeScript conventions                        |
| Month 2 (Hamle 2018 EC)    | Manual payment confirmation workflow implementation (Team 6), super admin dashboard (Team 2), invoice and receipt workspace, first cross-team integration  |
| Month 3 (Nehase 2018 EC)   | OCR-based screenshot verification pipeline, payment status normalization, org governance interfaces, Cursor Hackathon participation at Haramaya University |
| Month 4 (Meskerem 2019 EC) | Integration testing with Team 7, presentation preparation, documentation handover, ongoing client project collaboration                                    |

### Appendix B: Certificate of Participation

This appendix contains the certificate of participation awarded for the Cursor Hackathon held at Haramaya University during the internship period. The certificate recognises participation in the competitive software development event alongside students from multiple universities across Ethiopia.

[Insert certificate image here — the certificate is a scanned document that should be embedded at this location in the final printed report.]